

• Notes for Lectures 11 and 12

§ 11.1 Introduction

Definition 1. A positive Borel measure μ on $\mathbb{R}_+^{d+1} := \{(x, t) : x \in \mathbb{R}^d, t > 0\}$ is said to be a Carleson measure if there exists a constant $C_\mu > 0$ such that

$$\mu(Q \times (0, \ell(Q))) \leq C_\mu |Q|$$

for every cube Q in $\mathbb{R}^d$. Here, $\ell(Q)$ denotes the length of Q .

If μ is a Carleson measure, then its corresponding Carleson constant is defined by

$$\|\mu\| := \sup_{\substack{Q \subseteq \mathbb{R}^d \\ \text{cube}}} \frac{\mu(Q \times (0, \ell(Q)))}{|Q|}.$$

Notation: • If Q is a cube in $\mathbb{R}^d$, define the tent over Q to be the cube given by

$$T(Q) := Q \times (0, \ell(Q)).$$

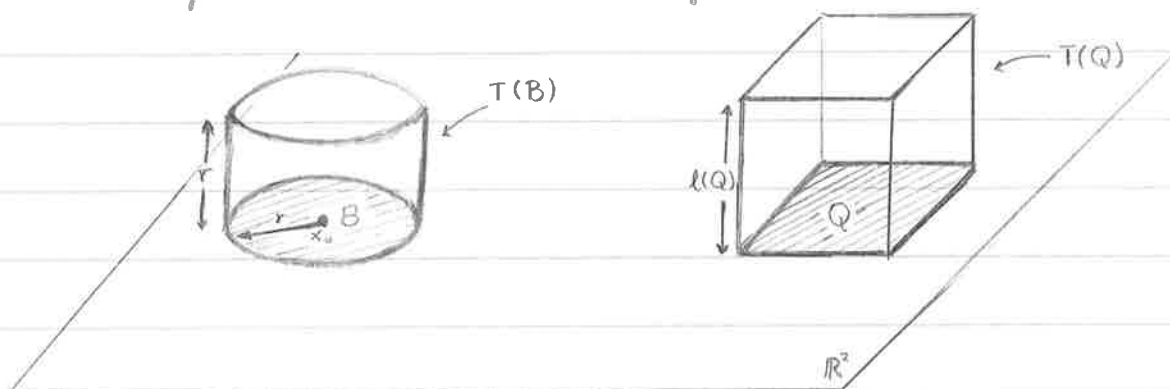
• If $B = B(x_0, r)$ is a ball in $\mathbb{R}^d$, define the cylindrical tent over B by

$$T(B) := \{(x, t) \in \mathbb{R}_+^{d+1} : x \in B, 0 < t \leq r\}.$$

Exercise. Let μ be a positive Borel measure on $\mathbb{R}_+^{d+1}$. Show that μ is a Carleson measure if, and only if,

$$\|\mu\|_{\text{cylinder}} := \sup_{\substack{B \subseteq \mathbb{R}^d \\ \text{ball}}} \frac{\mu(T(B))}{|B|} < \infty$$

and $\|\mu\|_{\text{cylinder}} \sim_d \|\mu\|$. (that is, there exist constants $C_d, C'_d > 0$, depending only on d , such that $C'_d \|\mu\|_{\text{cylinder}} \leq \|\mu\| \leq C_d \|\mu\|_{\text{cylinder}}$).



The main goal of this lecture is to prove a theorem of L. Carleson that connects Carleson measures with certain boundedness properties of operators defined in terms of approximate identities. To state L. Carleson's theorem, recall that for $t > 0$ and a function g on $\mathbb{R}^d$ we write $g_t(x) := t^{-d} g(t^{-1}x)$, $x \in \mathbb{R}^d$.

Theorem 2. Let Φ be a non-negative measurable function on $\mathbb{R}^d$ such that

$$\Phi(x) \leq \frac{C}{(1+|x|)^{d+\delta}} \quad \text{for a.e. } x \in \mathbb{R}^d,$$

where $C, \delta > 0$ are constants.

Let μ be a positive Borel measure on $\mathbb{R}_+^{d+1}$.

Then, the following are equivalent:

- (i) μ is a Carleson measure,
 (ii) For every $p \in (1, \infty)$ there exists a constant $C_{p,d,\mu} > 0$ such that

$$\int_{\mathbb{R}_+^{d+1}} |(\Phi_t * f)(x)|^p d\mu(x, t) \leq C_{p,d,\mu} \int_{\mathbb{R}^d} |f(x)|^p dx.$$

In order to establish Theorem 2, we need some auxiliary results that are of independent interest.

Theorem 3 (Whitney decomposition). Let Ω be an non-empty open set in $\mathbb{R}^d$ with $\Omega \subsetneq \mathbb{R}^d$.

Then, there exists a countable collection $\mathcal{Q}$ of closed cubes in $\mathbb{R}^d$ such that:

- (a) the cubes in $\mathcal{Q}$ have mutually disjoint interiors,
 (b) $\Omega = \bigcup_{Q \in \mathcal{Q}} Q$,

(c) $\sqrt{d} \cdot \ell(Q) \leq \text{dist}(Q, \mathbb{R}^d \setminus \Omega) \leq 4\sqrt{d} \cdot \ell(Q).$

Proof. See Appendix J in [2].

Proposition 4. Let μ be a positive Borel measure on $\mathbb{R}_+^{d+1}$.

The following are equivalent:

- (i) μ is a Carleson measure

(ii) There exists a constant $C_{\mu,d} > 0$ such that for every non-empty open set Ω in $\mathbb{R}^d$ with $\Omega \subsetneq \mathbb{R}^d$ one has

$$\mu(\{(x,t) \in \mathbb{R}_+^{d+1} : B(x,t) \subseteq \Omega\}) \leq C_{\mu,d} \cdot |\Omega|.$$

Proof. (i) $\Rightarrow$ (ii) Suppose that μ is a Carleson measure and let Ω be a non-empty set with $\Omega \subsetneq \mathbb{R}^d$.

By employing Theorem 3, there exists a countable collection of closed cubes Q in $\mathbb{R}^d$ such that:

- (1) $\Omega = \bigcup_{Q \in \mathcal{Q}} Q$ with the cubes in $\mathcal{Q}$ having mutually disjoint interiors,
- (2) $\sqrt{d} \ell(Q) \leq \text{dist}(Q, \mathbb{R}^d \setminus \Omega) \leq 4\sqrt{d} \ell(Q)$ for all $Q \in \mathcal{Q}$

Write $\tilde{\Omega} := \{(x,t) \in \mathbb{R}_+^{d+1} : B(x,t) \subseteq \Omega\}$.

Claim. 'If $\Omega, \mathcal{Q} = \{Q\}_{Q \in \mathcal{Q}}$ are as above, then

$$\tilde{\Omega} \subseteq \bigcup_{Q \in \mathcal{Q}} Q \times (0, 11\sqrt{d} \ell(Q)).'$$

Assume that the claim holds, then

$$\begin{aligned} \mu(\tilde{\Omega}) &\leq \mu\left(\bigcup_{Q \in \mathcal{Q}} Q \times (0, 11\sqrt{d} \ell(Q))\right) \\ &\leq \sum_{Q \in \mathcal{Q}} \mu(Q \times (0, 11\sqrt{d} \ell(Q))) \\ &\leq \sum_{Q \in \mathcal{Q}} \mu(Q^* \times (0, \ell(Q^*))), \end{aligned}$$

where Q^* is the dilation of Q with length equal to

$l(Q^*) = 11\sqrt{d} \cdot l(Q)$. Hence, the previous estimate yields

$$\mu(\tilde{\Omega}) \leq \|\mu\| \cdot \sum_{Q \in \mathcal{Q}} |Q^*|$$

$$\text{and so, } \mu(\tilde{\Omega}) \leq 11^d \cdot d^{d/2} \cdot \|\mu\| \cdot \sum_{Q \in \mathcal{Q}} |Q|$$

$$= 11^d \cdot d^{d/2} \cdot \|\mu\| \cdot \left| \bigcup_{Q \in \mathcal{Q}} Q \right| \quad \left(\begin{array}{l} \text{the cubes in } \mathcal{Q} \\ \text{have mutually disjoint} \\ \text{interiors} \end{array} \right)$$

$$= 11^d \cdot d^{d/2} \cdot \|\mu\| \cdot |\Omega| \quad (\text{by (1)}).$$

Therefore, (ii) holds. It remains to establish the aforementioned claim.

Proof (of the claim). Let $(x, t) \in \tilde{\Omega}$. Then $x \in \Omega$ and so, there exists a $Q \in \mathcal{Q}$ such that $x \in Q$. Since Q and $\mathbb{R}^d \setminus \Omega$ are closed, there exists a $y_0 \in \mathbb{R}^d \setminus \Omega$ and an $x_0 \in Q$ such that

$$(3) \quad \text{dist}(Q, \mathbb{R}^d \setminus \Omega) = |x_0 - y_0|.$$

Consider the open ball

$$B_0 := B(y_0, 6\sqrt{d}l(Q)).$$

We claim that

$$(4) \quad Q \subseteq B_0.$$

Indeed, if x_Q denotes the centre of Q , then for every $x \in Q$ one has

$$\begin{aligned} |x - y_0| &\leq |x - x_Q| + |x_Q - x_0| + |x_0 - y_0| \\ &\leq \sqrt{d} \cdot \frac{l(Q)}{2} + \sqrt{d} \cdot \frac{l(Q)}{2} + |x_0 - y_0| = \sqrt{d} \cdot l(Q) + |x_0 - y_0|, \end{aligned}$$

as $x_0, x \in Q$. By using (2) and (3), we get

$$|x - y_0| \leq \sqrt{d} \cdot l(Q) + 4\sqrt{d} \cdot l(Q) < 6\sqrt{d} \cdot l(Q).$$

Hence, (4) holds.

Observe that, by our construction, $B_0 \cap (\mathbb{R}^d \setminus \mathcal{Q}) \neq \emptyset$. This implies that $t < 11\sqrt{d} \ell(Q)$, where $t > 0$ is as above, i.e. $(x, t) \in \tilde{\mathcal{Q}}$ and so, $B(x, t) \subseteq \mathcal{Q}$. Indeed, to show that $t < 11\sqrt{d} \ell(Q)$, we argue by contradiction assuming that $t \geq 11\sqrt{d} \ell(Q)$. Then for every $y \in B_0$ one has

$$\begin{aligned} |y - x| &\leq |y - y_0| + |y_0 - x_0| + |x_0 - x| \\ &\leq 6\sqrt{d} \ell(Q) + 4\sqrt{d} \ell(Q) + \frac{\sqrt{d}}{2} \ell(Q) \\ &< 11\sqrt{d} \ell(Q) \leq t \end{aligned}$$

where y_0, x_0 are as in (3). Hence, $y \in B(x, t)$ and so, $B_0 \subseteq B(x, t) \subseteq \mathcal{Q}$, contradiction.

Therefore, $t < 11\sqrt{d} \ell(Q)$ and so,

$$\tilde{\mathcal{Q}} \subseteq \bigcup_{Q \in \mathcal{Q}} Q \times (0, 11\sqrt{d} \ell(Q)). \quad \square$$

Having completed the proof of the claim of page 4, the proof of the implication (i) $\Rightarrow$ (ii) is complete.

(ii) $\Rightarrow$ (i) Exercise. ■

Having established Proposition 4, we are almost ready to prove Theorem 2. Before doing so, let us the following remark.

Remark 5. Let Φ be as in the statement

of Theorem 2, then there exists a constant $A_{\Phi, d}$, depending on Φ and d , such that for $f \in L^p(\mathbb{R}^d)$, $1 \leq p < \infty$, one has

$$(5) \quad M_{\Phi}(f)(x) := \sup_{(y, t) \in \Gamma(x)} |\Phi_t * f(y)| \leq A_{\Phi, d} \cdot M(f)(x)$$

for a.e. $x \in \mathbb{R}^d$, where M is the Hardy-Littlewood operator and

$$\Gamma(x) := \{(y, t) \in \mathbb{R}_+^{d+1} : |x - y| < t\}$$

is the cone with vertex x truncated at height t .

For a proof of (5), see Exercise 3.3.4 in [3].

§ 11.2 Proof of Theorem 2

(i) $\Rightarrow$ (ii) Suppose that μ is a Carleson measure. Let $p \in (1, \infty)$ and let $f \in L^p(\mathbb{R}^d)$ be given.

We have

$$\int_{\mathbb{R}_+^{d+1}} |\Phi_t * f(x)|^p d\mu(x, t) = \int_0^\infty p \lambda^{p-1} \mu(\{(x, t) \in \mathbb{R}_+^{d+1} : |\Phi_t * f(x)| > \lambda\}) d\lambda.$$

Write $\Omega_\lambda := \{x \in \mathbb{R}^d : M_{\Phi}(f)(x) > \lambda\}$, $\lambda > 0$.

Exercise. For $\lambda > 0$, show that Ω_λ is open.

By using the previous exercise and Proposition 4 and the fact that, for all $\lambda > 0$, one has

$$\{(x, t) \in \mathbb{R}_+^{d+1} : |\Phi_t * f(x)| > \lambda\} \subseteq \widetilde{\Omega}_\lambda \quad (\text{as}$$

$M_{\Phi}(f)(x) = \sup_{(y, t) \in \Gamma(x)} |\Phi_t * f(y)| \geq |\Phi_t * f(x)|$), we deduce

$\mu(\{(x,t) \in \mathbb{R}_+^{d+1} : |\Phi_t * f(x)| > \lambda\}) \leq \mu(\tilde{\Omega}_\lambda) \leq C_{\mu,d} \cdot |\Omega_\lambda|$
 for all $\lambda > 0$, where $C_{\mu,d} > 0$ depends only on μ and d and is independent of f, λ .

We thus have

$$\begin{aligned} \int_{\mathbb{R}_+^{d+1}} |\Phi_t * f(x)|^p d\mu(x,t) &\leq C_{\mu,d} \cdot \int_0^\infty p \cdot \lambda^{p-1} \cdot |\{x \in \mathbb{R}^d : M_\Phi(f)(x) > \lambda\}| d\lambda \\ &= C_{\mu,d} \cdot \int_{\mathbb{R}^d} (M_\Phi(f)(x))^p dx \\ &\stackrel{(5)}{\leq} C_{\mu,d} \cdot A_{\Phi,d}^p \cdot \int_{\mathbb{R}^d} (M(f)(x))^p dx \\ &\leq C_{\mu,d} \cdot A_{\Phi,d}^p \cdot C_d \cdot \frac{p}{p-1} \cdot \int_{\mathbb{R}^d} |f(x)|^p dx, \end{aligned}$$

where in the last we used the fact that the Hardy-Littlewood maximal operator is bounded on $L^p(\mathbb{R}^d)$ with $\|M\|_{L^p(\mathbb{R}^d) \rightarrow L^p(\mathbb{R}^d)} \leq C_d \cdot \frac{p}{p-1}$.

Hence, the proof of the implication (i) $\Rightarrow$ (ii) is complete.

(ii) $\Rightarrow$ (i) Suppose that μ is a positive Borel measure on $\mathbb{R}_+^{d+1}$ such that (ii) holds.

Let $B_0 := B(x_0, r)$ be an open ball in $\mathbb{R}^d$.
 Let $(x,t) \in \mathbb{R}_+^{d+1}$ be such that $B(x,t) \subseteq B_0$. Then

$$|\Phi_t * \chi_{2B_0}(x)| = \int_{2B_0} \Phi_t(x-y) dy \geq \int_{x-2B_0} \Phi_t(y) dy.$$

Since $B(x,t) \subseteq B_0$, we have $x-2B_0 \supseteq B(0,t)$ and

So,

$$|\bar{\Phi}_t * \chi_{2B_0}(x)| \geq \int_{B(0,t)} \Phi_t(y) dy = \int_{B(0,1)} \Phi(y) dy = C_{d,\Phi} > 0.$$

Hence,

$$\mu(\{(x,t) \in \mathbb{R}_+^{d+1} : B(x,t) \subseteq B_0\}) \leq$$

$$\frac{1}{(C_{d,\Phi})^p} \int_{\mathbb{R}_+^{d+1}} |\bar{\Phi}_t * \chi_{2B_0}(x)|^p d\mu(x,t) \stackrel{(ii)}{\leq}$$

$$\left(\frac{C_{p,d,\mu}}{C_{d,\Phi}}\right)^p \int_{\mathbb{R}^d} |\chi_{2B_0}(x)|^p dx \lesssim_{p,d,\mu,\Phi} |B_0|.$$

Hence, by Proposition 4, μ is a Carleson measure. ■

Corollary 6. Let μ be a positive Borel measure on $\mathbb{R}_+^{d+1}$.

The following are equivalent:

- (i) μ is a Carleson measure.
- (ii) For every $p \in (1, \infty)$ there exists a constant $C_{p,d,\mu} > 0$ such that

$$\left(\int_{\mathbb{R}_+^{d+1}} |(P_t * f)(x)|^p d\mu(x,t) \right)^{1/p} \leq C_{p,d,\mu} \cdot \|f\|_{L^p(\mathbb{R}^d)}.$$

Here, $(P_t)_{t>0}$ is the Poisson kernel on $\mathbb{R}^d$,

$$P_t(x) := \frac{\Gamma((d+1)/2)}{\pi^{(d+1)/2}} \cdot \frac{t}{(t^2 + |x|^2)^{(d+1)/2}},$$

$$x \in \mathbb{R}^d, \quad t > 0.$$

§ 12.1 Carleson measures and BMO

Theorem 7. Let b be a function of bounded mean oscillation on $\mathbb{R}^d$.

Let $\psi \in \mathcal{S}(\mathbb{R}^d)$ be such that

$$\hat{\psi}(0) = \int_{\mathbb{R}^d} \psi(y) dy = 0.$$

Then the measure μ on $\mathbb{R}_+^{d+1}$ given by

$$d\mu(x, t) := |b * \psi_t(x)|^2 dx \frac{dt}{t}$$

is a Carleson measure and moreover, there exists a constant $C_{d, \psi} > 0$, depending on d and ψ , such that

$$\|\mu\| \leq C_d \cdot \|b\|_{\text{BMO}(\mathbb{R}^d)}^2.$$

To establish Theorem 7 we need the following two lemmas that are of independent interest.

Lemma 8. Let b be a function of bounded mean oscillation on $\mathbb{R}^d$.

Let $Q \subseteq \mathbb{R}^d$ be a cube and for $N \in \mathbb{N}$, let $2^N Q$ denote the dilation of Q such that $\ell(2^N Q) = 2^N \ell(Q)$.

Then,

$$|\langle b \rangle_Q - \langle b \rangle_{2^N Q}| \leq 2^d \cdot N \cdot \|b\|_{\text{BMO}(\mathbb{R}^d)}.$$

Proof. Let Q_0 be a cube in $\mathbb{R}^d$. We have

$$\begin{aligned}
 |\langle b \rangle_{Q_0} - \langle b \rangle_{2Q_0}| &= \frac{1}{|Q_0|} \left| \int_{Q_0} (b(x) - \langle b \rangle_{2Q_0}) dx \right| \\
 &\leq \frac{1}{|Q_0|} \int_{Q_0} |b(x) - \langle b \rangle_{2Q_0}| dx \\
 &\leq \frac{1}{|Q_0|} \int_{2Q_0} |b(x) - \langle b \rangle_{2Q_0}| dx \\
 &= \frac{2^d}{|2Q_0|} \int_{2Q_0} |b(x) - \langle b \rangle_{2Q_0}| dx \\
 &\leq 2^d \cdot \|b\|_{BMO(\mathbb{R}^d)},
 \end{aligned}$$

where $2Q_0$ is a dilation of Q_0 with $\ell(2Q_0) = 2\ell(Q_0)$. We thus have

$$(6) \quad |\langle b \rangle_{Q_0} - \langle b \rangle_{2Q_0}| \leq 2^d \cdot \|b\|_{BMO(\mathbb{R}^d)}.$$

We have

$$\begin{aligned}
 |\langle b \rangle_Q - \langle b \rangle_{2^N Q}| &= |\langle b \rangle_Q - \langle b \rangle_{2Q} + \dots + \langle b \rangle_{2^{N-1}Q} - \langle b \rangle_{2^N Q}| \\
 &\leq |\langle b \rangle_Q - \langle b \rangle_{2Q}| + \dots + |\langle b \rangle_{2^{N-1}Q} - \langle b \rangle_{2^N Q}| \\
 &\stackrel{(6)}{\leq} 2^d \cdot N \cdot \|b\|_{BMO(\mathbb{R}^d)}.
 \end{aligned}$$

Lemma 9. Let b be a function of bounded mean oscillation on $\mathbb{R}^d$.

Let $\delta > 0$ be given. There exists a constant $C_{d,\delta} > 0$ such that for any cube Q one has

$$[\ell(Q)]^\delta \cdot \int_{\mathbb{R}^d} \frac{|b(x) - \langle b \rangle_Q|}{[\ell(Q) + |x - c_Q|]^{d+\delta}} dx \leq C_{d,\delta} \cdot \|b\|_{BMO(\mathbb{R}^d)}.$$

where $l(Q)$ denotes the length of Q and c_Q denotes the centre of Q .

Proof. Assume first that Q is the unit closed cube in $\mathbb{R}^d$ i.e. $Q = [-\frac{1}{2}, \frac{1}{2}]^d$.

We have

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{|b(x) - \langle b \rangle_Q|}{(1 + |x|)^{d+\delta}} dx &= \int_Q \frac{|b(x) - \langle b \rangle_Q|}{(1 + |x|)^{d+\delta}} dx \\ &+ \sum_{N=0}^{\infty} \int_{2^{N+1}Q \setminus (2^N Q)} \frac{|b(x) - \langle b \rangle_Q|}{(1 + |x|)^{d+\delta}} dx. \end{aligned}$$

We shall treat each of the terms on the right-hand side of the equality separately.

For the first term we have

$$\begin{aligned} A &:= \int_Q \frac{|b(x) - \langle b \rangle_Q|}{(1 + |x|)^{d+\delta}} dx \leq \int_Q |b(x) - \langle b \rangle_Q| dx \\ &\stackrel{|Q|=1}{=} \frac{1}{|Q|} \int_Q |b(x) - \langle b \rangle_Q| dx \\ &\leq \|b\|_{\text{BMO}(\mathbb{R}^d)}. \end{aligned}$$

For the second term we shall use Lemma 8 as follows,

$$\begin{aligned} B &:= \sum_{N=0}^{\infty} \int_{2^{N+1}Q \setminus (2^N Q)} \frac{|b(x) - \langle b \rangle_Q|}{(1 + |x|)^{d+\delta}} dx \\ &\leq \sum_{N=0}^{\infty} \int_{2^{N+1}Q \setminus (2^N Q)} \frac{|b(x) - \langle b \rangle_{2^{N+1}Q}| + |\langle b \rangle_{2^{N+1}Q} - \langle b \rangle_Q|}{(1 + |x|)^{d+\delta}} dx \end{aligned}$$

Note that if $x \notin 2^N Q$ then $|x| \geq 2^{N-1}$. Hence,

$$B \leq \sum_{N=0}^{\infty} \int_{2^{N+1}Q \setminus (2^N Q)} \frac{|b(x) - \langle b \rangle_{2^{N+1}Q}| + |\langle b \rangle_{2^{N+1}Q} - \langle b \rangle_Q|}{(1 + 2^{N-1})^{d+\delta}} dx$$

$$\leq \sum_{N=0}^{\infty} (1 + 2^{N-1})^{-(d+\delta)} \left(\int_{2^{N+1}Q} |b(x) - \langle b \rangle_{2^{N+1}Q}| dx + |2^{N+1}Q| \cdot |\langle b \rangle_{2^{N+1}Q} - \langle b \rangle_Q| \right)$$

$$\stackrel{\text{Lemma 8}}{\leq} \sum_{N=0}^{\infty} (1 + 2^{N-1})^{-(d+\delta)} \cdot \left(|2^{N+1}Q| \cdot \|b\|_{BMO(\mathbb{R}^d)} + |2^{N+1}Q| \cdot 2^d \cdot N \cdot \|b\|_{BMO(\mathbb{R}^d)} \right)$$

$$= \|b\|_{BMO(\mathbb{R}^d)} \cdot \sum_{N=0}^{\infty} (1 + 2^{N-1})^{-(d+\delta)} \cdot (2^{Nd} + N \cdot 2^d \cdot 2^{Nd})$$

$$\leq \|b\|_{BMO(\mathbb{R}^d)} \cdot \sum_{N=0}^{\infty} \frac{2^{d+\delta} (1 + 2^d \cdot N)}{2^{N\delta}}$$

$$\leq C_{d,\delta} \cdot \|b\|_{BMO(\mathbb{R}^d)},$$

$$\text{where } C_{d,\delta} = 2^{2d+\delta+1} \cdot \sum_{N=0}^{\infty} N \cdot 2^{-N\delta} < \infty.$$

The general case (i.e. for general cubes) is left as an exercise. ■

Proof (of Theorem 7). Let Q be a cube in $\mathbb{R}^d$.

We shall prove that

$$\mu(Q \times (0, l(Q))) \lesssim_{d, \psi} \|b\|_{BMO(\mathbb{R}^d)}^2 |Q|.$$

Set $f(x, t) := \chi_{Q \times (0, l(Q))}(x, t)$, $(x, t) \in \mathbb{R}_+^{d+1}$.

Then

$$\begin{aligned} \mu(Q \times (0, l(Q))) &= \int_{\mathbb{R}_+^{d+1}} f(x, t) d\mu(x, t) \\ &= \int_Q \left(\int_{(0, l(Q))} |b * \psi_t(x)|^2 \frac{dt}{t} \right) dx \end{aligned}$$

Let Q^* denote the dilation of Q with $l(Q^*) = 3\sqrt{d} l(Q)$.

Write

$$b = (b - \langle b \rangle_Q) \chi_{Q^*} + (b - \langle b \rangle_Q) \chi_{\mathbb{R}^d \setminus Q^*} + \langle b \rangle_Q.$$

Since $\int_{\mathbb{R}^d} \psi(y) dy = 0$,

$$b * \psi_t(x) = (b - \langle b \rangle_Q) \chi_{Q^*} * \psi_t(x)$$

$$+ (b - \langle b \rangle_Q) \chi_{\mathbb{R}^d \setminus Q^*} * \psi_t(x)$$

and so, by using the inequality $(a+b)^2 \leq 2a^2 + 2b^2$, we get

$$\mu(Q \times (0, l(Q))) \leq 2A + 2B,$$

where

$$A := \int_Q \left(\int_{(0, l(Q))} |(b - \langle b \rangle_Q) \chi_{Q^*} * \psi_t(x)|^2 \frac{dt}{t} \right) dx$$

and

$$B := \int_Q \left(\int_{(0, l(Q))} |(b - \langle b \rangle_Q) \chi_{\mathbb{R}^d \setminus Q^*} * \psi_t(x)|^2 \frac{dt}{t} \right) dx.$$

To handle A , we use Plancherel's theorem (in the x variable),

$$A \leq \int_{\mathbb{R}^d} \left(\int_{(0, \ell(Q))} |(b - \langle b \rangle_Q) \cdot \chi_{Q^*} * \psi_t(x)|^2 \frac{dt}{t} \right) dx$$

$$= \int_{\mathbb{R}^d} \left(\int_{(0, \ell(Q))} |\widehat{(b - \langle b \rangle_Q) \chi_{Q^*}}(\xi)|^2 |\widehat{\psi}(t\xi)|^2 \frac{dt}{t} \right) d\xi$$

$$\leq \int_{\mathbb{R}^d} \left(\int_{(0, \infty)} |\widehat{(b - \langle b \rangle_Q) \chi_{Q^*}}(\xi)|^2 |\widehat{\psi}(t\xi)|^2 \frac{dt}{t} \right) d\xi$$

Exercise. Show that if $\psi \in \mathcal{S}(\mathbb{R}^d)$ and $\widehat{\psi}(0) = \int_{\mathbb{R}^d} \psi(y) dy = 0$, then

$$|\widehat{\psi}(t\xi)| \lesssim_{\psi} \min \{ |t\xi|, |t\xi|^{-1} \}$$

for all $\xi \in \mathbb{R}^d$ and $t > 0$.

Using the previous exercise and Tonelli's theorem, we have

$$A \leq \int_{\mathbb{R}^d} |\widehat{(b - \langle b \rangle_Q) \chi_{Q^*}}(\xi)|^2 \left(\int_{(0, \infty)} |\widehat{\psi}(t\xi)|^2 \frac{dt}{t} \right) d\xi$$

$$\lesssim_{\psi} \int_{\mathbb{R}^d} |\widehat{(b - \langle b \rangle_Q) \chi_{Q^*}}(\xi)|^2 \left(\int_0^{1/|\xi|} |t\xi|^2 \frac{dt}{t} + \int_{1/|\xi|}^{\infty} \frac{1}{|t\xi|^2} \frac{dt}{t} \right) d\xi$$

$$\lesssim \int_{\mathbb{R}^d} |\widehat{(b - \langle b \rangle_Q) \chi_{Q^*}}(\xi)|^2 d\xi$$

and so, by using Plancherel's theorem, (a variant of) Lemma 8, and Corollary 4 of Lecture 10, we get

$$A \lesssim_{\psi} \int_{Q^*} |b(x) - \langle b \rangle_Q|^2 dx$$

$$\leq \int_{Q^*} |b(x) - \langle b \rangle_{Q^*}|^2 dx + |Q^*| |\langle b \rangle_Q - \langle b \rangle_{Q^*}|^2$$

$$\lesssim |Q| \cdot \|b\|_{BMO(\mathbb{R}^d)}^2.$$

To handle B, let $2^N Q$ denote the dilate of cube Q with $\ell(2^N Q) = 2^N \ell(Q)$. We have

$$B \leq \int_Q \int_{(0, \ell(Q))} \left| \sum_{N=0}^{\infty} \int_{2^{N+1}Q \setminus 2^N Q} (b(y) - \langle b \rangle_{Q^*}) \psi_t(x-y) dy \right|^2 \frac{dt}{t} dx$$

Exercise. Show that if $\psi \in \mathcal{S}(\mathbb{R}^d)$, $x \in Q$, then

$$\int_Q \int_{(0, \ell(Q))} \left| \int_{2^{N+1}Q \setminus 2^N Q} (b(y) - \langle b \rangle_{Q^*}) \psi_t(x-y) dy \right|^2 \frac{dt}{t} dx \lesssim_{\psi} |Q| \|b\|_{BMO(\mathbb{R}^d)}^2$$

By using the previous exercise, one deduces that

$$\mu(Q \times (0, \ell(Q))) \lesssim \|b\|_{BMO(\mathbb{R}^d)}^2 \cdot |Q|. \quad \blacksquare$$

In fact, Theorem 7 has a converse; see §4.3 - §4.4 in [4].

The notes of this lecture are based on the following references.

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